P101 · Unit 10 · Quick reference

Physics, space & scientific thinking
Light, images and observational limits
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Recall **Vacuum light**	$c \approx 3.00\times10^{8}\ \mathrm{m\,s^{-1}}$; $c = f\lambda_0$, $T = 1 / f$. Spectrum, increasing frequency: radio → microwave → infrared → visible → ultraviolet → X-ray → gamma ray.
Recall **Reflection**	$\theta_r = \theta_1$. Angles from the local normal; rays and normal coplanar. Surface angle $= 90^\circ - \theta_1$.
Recall **Refraction**	$n = c / v$; $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Stationary boundary: frequency fixed, $\lambda = \lambda_0 / n$. Higher index: smaller speed and normal angle.
Understand **Critical angle**	For $n_1 > n_2$: $\theta_c = \sin^{-1}(n_2 / n_1)$. Above it: total internal reflection at an ideal transparent lossless interface. Inverse sine is not reciprocal sine.
Recall **Thin converging lens**	$1 / f = 1 / d_o + 1 / d_i$; $f > 0, d_o > 0$. Real image beyond lens: $d_i > 0$; virtual, object side: $d_i < 0$. Thin, paraxial lens in air; common length units.
Understand **Image geometry**	Parallel ray → far focus; central ray → straight; near-focus ray → parallel. $m = h_i / h_o = -d_i / d_o$. Negative m: inverted. $d_o = f$: no finite image plane.

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Recall **Intensity and energy**	$I = P / (4\pi r^2)$ / W m⁻². Steady isotropic small source, $r > 0$, negligible losses. Small perpendicular collector: $P_{\text{received}} = IA$, $E = IA\Delta t$ for steady intensity.
Understand **Distance ratio**	Unchanged source/conditions: $I_2 / I_1 = (r_1 / r_2)^2$. Source power differs from received power. $1\ \text{cm}^2 = 10^{-4}\ \text{m}^2$.
Understand **Recorded values**	Matched additive background: source estimate $R - B$. Counts depend on exposure and response settings. Linear unsaturated response needed for proportional comparisons.
Understand **Observational limits**	Resolution: distinguish nearby features. Magnification: image/display size. Cadence: time between samples. Exposure: accumulation interval. Finer sampling cannot undo optical blur or exposure averaging.
Consult **Validation**	Check an absolute value, ratio, units, zero-power case and invalid inputs. Preserve raw records, settings and provenance. Hand agreement checks implementation; measured evidence tests the physical model.
Consult **Sampling comparison**	Compare full and sparse records on common axes; check sampling phase. Nearly equal samples do not establish a steady source between them. Repeated/control measurements test specified alternatives.

Consult · Python

Patterns use the Lesson 4 arrays raw, control and t_s.

```
import numpy as np
r = np.linspace(1, 5, 81)        # m; exclude zero
I = 12 / (4*np.pi*r**2)          # W/m²
# Matched background; arrays share times/settings
corrected = raw - control
sparse_t = t_s[::2]             # every second sample
sparse_y = corrected[::2]
span = np.max(corrected) - np.min(corrected)
```

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